

Low-power wide-area network

A **low-power, wide-area network** (**LPWAN** or **LPWA network**) is a type of wireless telecommunication wide area network designed to allow long-range communication at a low bit rate between IoT devices, such as sensors operated on a battery.

Low power, low bit rate, and intended use distinguish this type of network from a wireless WAN that is designed to connect users or businesses, and carry more data, using more power. The LPWAN data rate ranges from 0.3 kbit/s to 50 kbit/s per channel.

A LPWAN may be used to create a private wireless sensor network, but may also be a service or infrastructure offered by a third party, allowing the owners of sensors to deploy them in the field without investing in gateway technology.

Attributes

1. **Range:** The operating range of LPWAN technology varies from a few kilometers in urban areas to over 10 km in rural settings. It can also enable effective data communication in previously infeasible indoor and underground locations.
2. **Power:** LPWAN manufacturers claim years to decades of usable life from built-in batteries, but real-world application tests have not confirmed this.^[1]

Platforms and technologies

Some competing standards and vendors for LPWAN space include:^[2]

- DASH7, a low latency, bi-directional firmware standard that operates over multiple LPWAN radio technologies including LoRa.
- Wize is an open and royalty-free standard for LPWAN derived from the European Standard Wireless Mbus.^[3]
- Chirp spread spectrum (CSS) based devices.
- Sigfox, UNB-based technology and French company.^[4]
- LoRa is a proprietary, chirp spread spectrum radio modulation technology for LPWAN used by LoRaWAN, Haystack Technologies, and Symphony Link.^{[5][6]}
- MloTy, implementing Telegram Splitting technology.
- Weightless is an open standard, narrowband technology for LPWAN used by Ubiik (<http://www.ubiik.com>)
- ELTRES, a LPWA technology developed by Sony, with transmission ranges of over 100 km while moving at speeds of 100 km/h.^[7]
- IEEE 802.11ah, also known as Wi-Fi HaLow, is a low-power, wide-area implementation of 802.11 wireless networking standard using sub-gig frequencies.^[8]

Ultra-narrow band

Ultra Narrowband (UNB), modulation technology used for LPWAN by various companies including:

- Sigfox, French UNB-based technology company.^[9]
- Weightless, a set of communication standards from the Weightless SIG.^[10]
- NB-Fi Protocol, developed by WAVIoT company.^[11]

Others

- DASH7 Mode 2 development framework for low power wireless networks, by Haystack Technologies.^[12] Runs over many wireless radio standards like LoRa, LTE, 802.15.4g, and others.
- LTE Advanced for Machine Type Communications (LTE-M), an evolution of LTE communications for connected things by 3GPP.^[13]
- MySensors, DIY Home Automation framework supporting different radios including LoRa.
- NarrowBand IoT (NB-IoT), standardization effort by 3GPP for a LPWAN used in cellular networks.^[14]
- Random phase multiple access (RPMA) from Ingenu, formerly known as On-Ramp Wireless, is based on a variation of CDMA technology for cellular phones, but uses unlicensed 2.4GHz spectrum.^{[15][16]} RPMA is used in GE's AMI metering.^[17]
- Byron, a direct-sequence spread spectrum (DSSS) technology from Taggle Systems in Australia.^[18]
- Wi-SUN, based on IEEE 802.15.4g.^[19]

See also

- Internet of things
- Wide area networks
- Static Context Header Compression (SCHC)
- QRP operation
- Slowfeld
- Through-the-earth mine communications
- Short range device
- IEEE 802.15.4 (Low-power personal-area network)
- IEEE 802.16 (WiMAX)

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